EST 5053 · CCBS 2026 · LECTURE 10

The adaptation biomachine

Why integral feedback produces perfect adaptation

Weihan Luo

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Section 2 · An integral rests only at zero error

THE KEY QUESTION

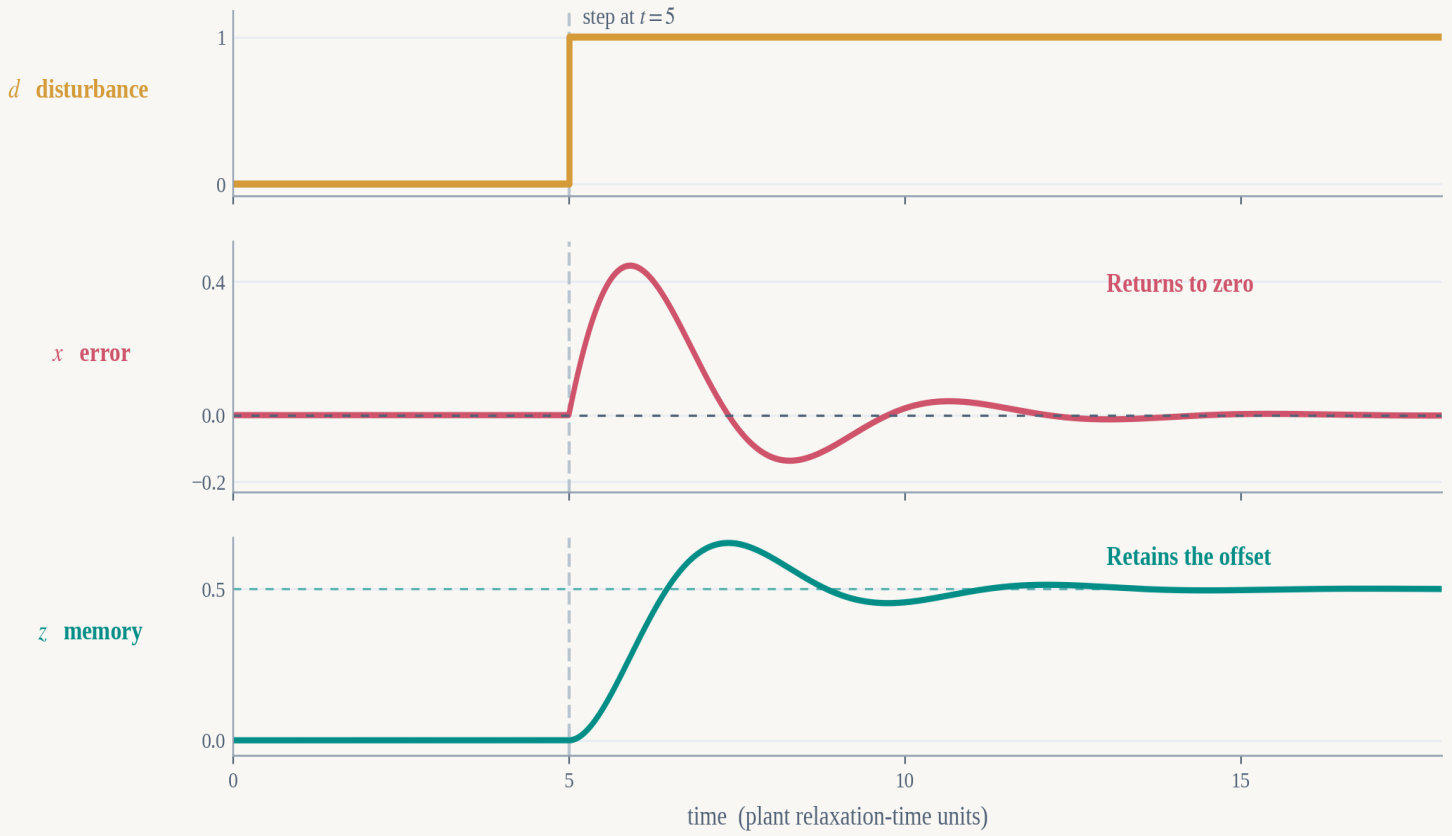
Where does the disturbance go?

The output returns to its set point, even though the disturbance stays on.

Disturbance persists...

...but no persistent error.

If the output comes back, something else must hold the disturbance.



The mathematical problem

Goal: design a memory that holds the disturbance at equilibrium, and prove the error goes to zero

$$\dot{x} = \underbrace{-x}_{\text{intrinsic relaxation (e.g. degradation)}} + u + d$$

- x output error
- u control action
- d persistent disturbance

Without control (u = 0)

$$\dot{x} = 0 \quad \Longrightarrow \quad x^* = d$$

The disturbance leaves a steady offset.

Our design goal: choose u(t)

$$\lim_{t \to \infty} x(t) = 0$$

for constant, nonzero disturbance d.

1 EQUILIBRIUM ANALYSIS → **2 STABILITY ANALYSIS**

Find the zero-error equilibrium. — *Prove we actually converge to it.*

Our proposed solution: integral feedback

The controller remembers accumulated error rather than only reacting to current error.

$$\dot{z} = x$$

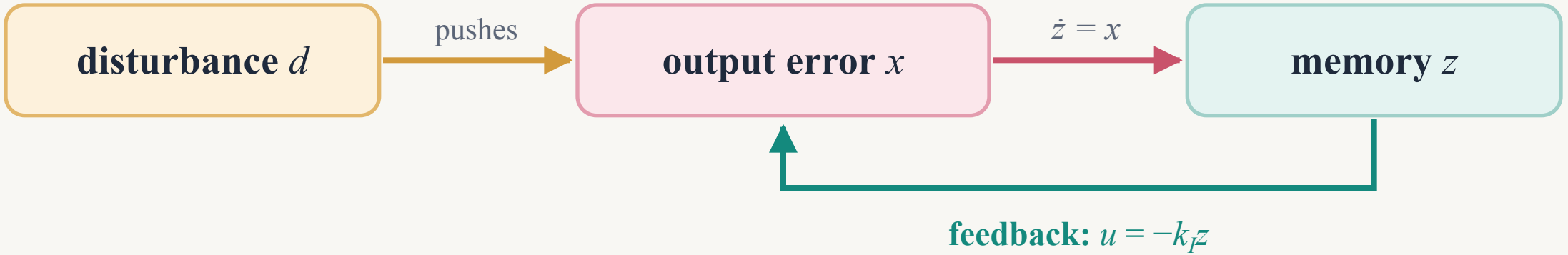
memory update

$$u = -k_I z$$

control action

z memory: the total error accumulated so far, $\int x\, dt$

$-k_I$ integral gain > 0; the minus sign makes the correction u oppose z



When x = 0, z stops changing; it need not become zero.

Lecture Figure 3, upper panel (redrawn)

CORE PROOF

Equilibrium analysis: zero steady-state error

If equilibrium is reached under this design, do we get zero error?

Closed-loop equations

$$\dot{x} = -x + d - k_I z$$

$$\dot{z} = x$$

1 The memory is stationary

$$0 = \dot{z} = x^* \implies \boxed{x^* = 0}$$

2 Balance the plant

$$0 = -x^* + d - k_I z^* \implies \boxed{z^* = \frac{d}{k_I}}$$

At equilibrium: the error is zero, and the memory holds the compensating action.

This proves where the equilibrium is — not yet that the system reaches it.

Stability analysis I: derive error dynamics

1. Differentiate the plant equation

$$\ddot{x} = -\dot{x} + \dot{d} - k_I \dot{z}$$

2. Use the constant-input assumption and controller

$$\dot{d} = 0,\ \dot{z} = x \quad \Longrightarrow \quad \ddot{x} = -\dot{x} - k_I x$$

3. **Rearrange into a damped second-order ODE**

$$\boxed{\ddot{x} + \dot{x} + k_I x = 0}$$

Stability analysis II: solve second-order ODE

Solutions decay when every characteristic root has negative real part.

$$x(t) \propto e^{rt} \quad \Longrightarrow \quad r^2 + r + k_I = 0$$

Solve the quadratic:

$$r_{\pm} = \frac{-1 \pm \sqrt{1 - 4k_I}}{2}$$

For the lecture's example,

$$k_I = 2 \quad \Longrightarrow \quad r_{\pm} = -\frac{1}{2} \pm i\frac{\sqrt{7}}{2}$$

Stability analysis III: why the roots prove convergence

Every solution is a sum of exponentials, and each one decays.

1. Every solution combines the two exponential modes

$$x(t) = A\, e^{r_+ t} + B\, e^{r_- t}$$

A, B set by the starting values

2. Only the real part of r sets the size of each mode

$$r = a + ib \implies |e^{rt}| = e^{at} \cdot |e^{ibt}| = e^{at}$$

Euler: $e^{ibt} = \cos bt + i \sin bt$, so $|e^{ibt}| = \sqrt{(\cos^2 bt + \sin^2 bt)} = 1$

3. Both roots have a negative real part for any positive gain

$$0 < k_I \le \tfrac{1}{4}: \quad -1 \pm \sqrt{(1 - 4k_I)} < 0 \qquad | \qquad k_I > \tfrac{1}{4}: \quad \mathrm{Re}(r_\pm) = -\tfrac{1}{2}$$

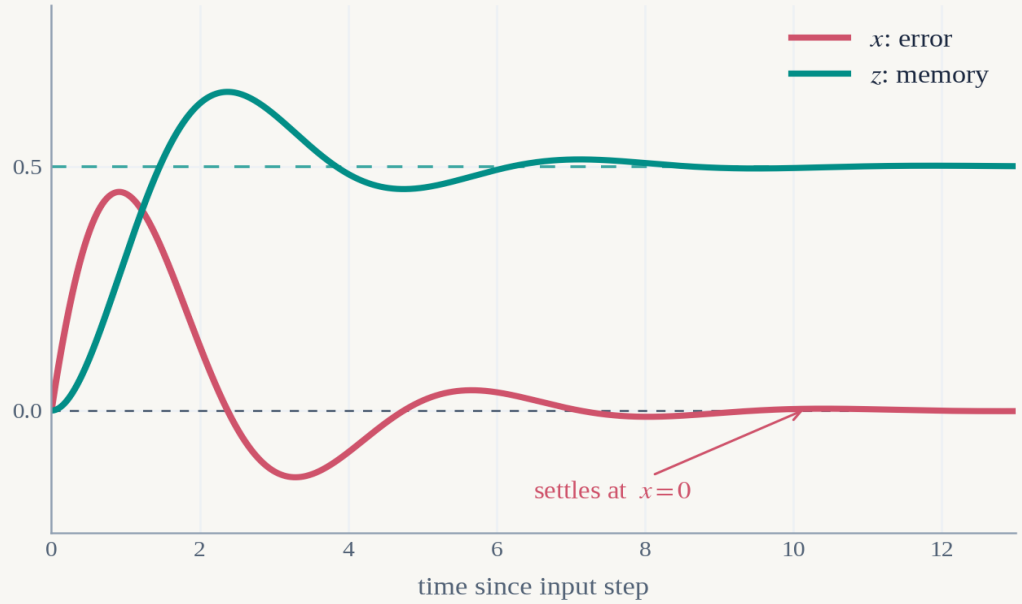
$$x(t) \to 0 \quad \text{and then} \quad z(t) \to d / k_I$$

Once x and its derivative vanish, the plant equation forces the memory to the slide-5 equilibrium.

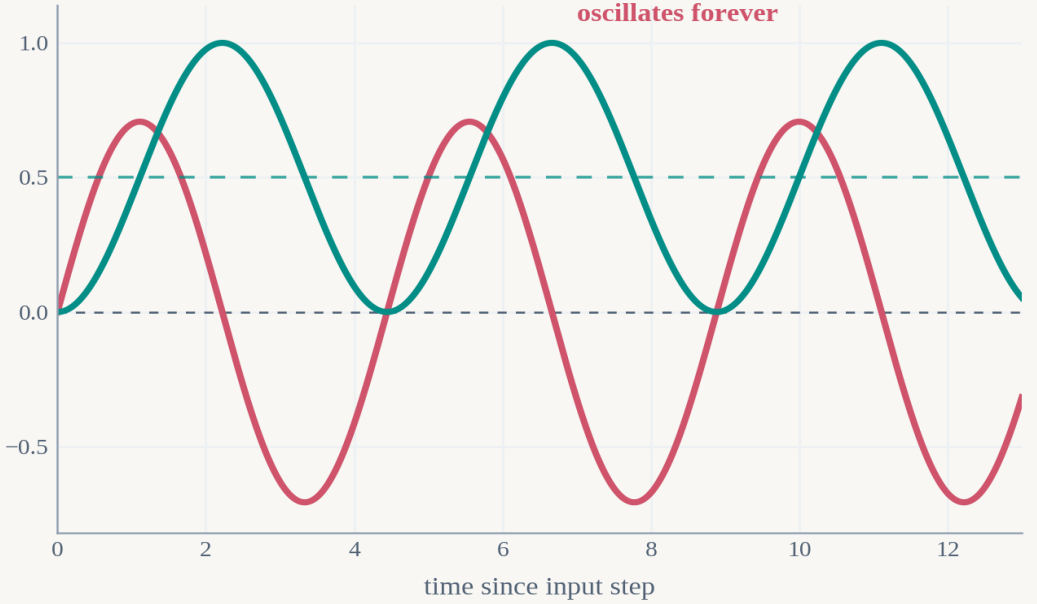
Relaxation provides the damping

Without the plant's intrinsic relaxation, the same integral controller oscillates.

WITH intrinsic relaxation $-x$ $r_{\pm} = -\tfrac{1}{2} \pm i\sqrt{7}/2$



WITHOUT intrinsic relaxation $r_{\pm} = 0 \pm i\sqrt{2}$



With −x: x(t) returns to zero

$$\dot{x} = -x + d - k_I z, \quad \dot{z} = x$$

Without −x: oscillations persist

$$\dot{x} = d - k_I z, \quad \dot{z} = x$$

Lecture Figure 4 (redrawn)

Why integral feedback?

Same plant, two controllers: only one reaches zero error.

$$\dot{x} = -x + u + d$$

Proportional feedback

$$u = -k_P x$$

$$\Longrightarrow \; x^* = d \,/\, (1 + k_P)$$

Higher gain shrinks the error, but never removes it.

Integral feedback

$$\dot{z} = x, \quad u = -k_I z$$

$$\Longrightarrow \; x^* = 0$$

True for any disturbance, gain, or plant constants.

To push back with zero error, the controller needs memory.

Internal model principle (Francis & Wonham, 1976): rejecting constant disturbances robustly requires an integrator.

TAKEAWAY

So, where does the disturbance go?

Integral feedback stores the disturbance in memory, so the error returns to zero.